batmonitor 0.59-c: one cycle every 5 s | blue = decision (bits + state, make check) | green = matrix / amount | details: charge_flow.pdf, discharge_flow.pdf

inputs (bm_parse: only valid numbers, else stale)
PCC, Sofar Bat1, PV, heat pump W, R290 Hz, BMS SoC / U / CCL / DCL / lowest cell

D5 PCC > 20 kW for 2 min (at once > 23 kW) -> DO4 pulse
pv_relay/DO4 (FoxESS shedding)
first: R290 boost + air conditioning at ≥ 18 kW
grid offline: AC-out feed > 5 kW / unit for 10 s -> at once

D1 B1FIRST, WPLOW (bm_d1, 0.59-c)
-> S3 source DAY / B1 / STK / WPG
below WP_SOC_MIN 21 %: the discharge leaves the WP to the grid
make check: 192 combinations; details wp_matrix.pdf

S0 caps per stack: chg_cap, dis_cap (charger, BMS CCL / DCL)

block

SoC

D2 mode per stack, first match:
BMS_MISSING > FORCE > STALE > LADESPERRE > CHARGE / FULL > PROT > CHARGING > DIS
FORCE = SoC < SOC_FORCE or LOW (DCL 0 / cell < 3.05 V; ends at SOC_MIN_RELEASE, 0.55-c)
make check: 16128 combinations per stack (84 bit sets x 12 global x 16 states) -> mode:
BMS_MISSING 8064 · FORCE 5760 · STALE 1152 · LADESPERRE 288 · CHARGE 216 ·
PROT 216 · CHARGING 180 · DIS 180 · FULL 72

D-lead: stack more than SOC_BALANCE_ON ahead

src, WPLOW

mode matrix BM_MODE_RULE
0 W | FORCE | charge | discharge

CHARGE

FORCE

DIS

charge path

D4 forecast export cap (May-Aug)
while the forecast expects export > 18 kW:
wait below the cap, charge only above it

charge amount
max(0, own + KP x (PCC + Bat1_eff - target))
target + EXPORT_CAP while D4 cap;
≤ sum of chg_cap

FORCE: min(500 W, chg_cap)
any time of day, no TOU window

alloc_levels up to chg_cap
stack by stack in CHARGE_PRIORITY,
lead stack last

0 W rows

discharge branch

BM_SRC_RULE[src]
STK: Bat1 signed, +trickle, idle 0
else: Bat1 charge only, idle 20 W

D3 PROP / IDLE

discharge amount
min(PROP ? deficit : idle, W_MAX)
house = PCC + own charge (+ WP while WPLOW)

alloc_levels up to dis_cap
lead stack first, else
every stack the same share

setpoints per phase -> Multis (+ charge / - discharge)
rule text: mode SRC: split [CAP] [CCL / DCL] [WPLOW]
-> state.json -> victron2db.py -> wagodb.pv_victron.bm_l1..3_rule (every minute)